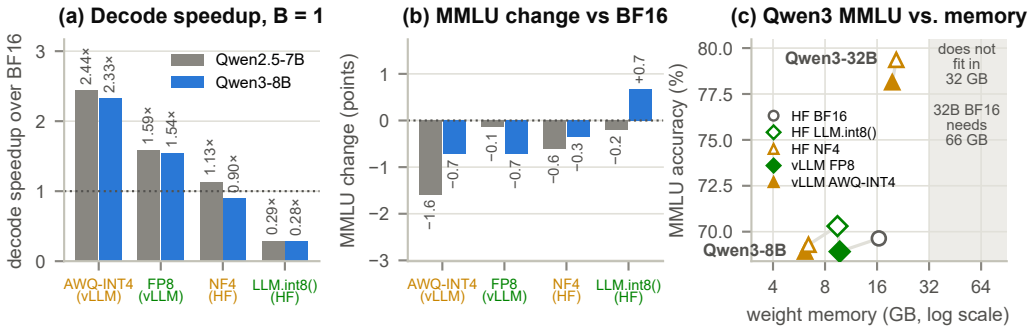
(a) Decode speedup, B = 1
decode speedup over BF16
Qwen2.5-7B
Qwen3-8B
2.44×
2.33×
1.59×
1.54×
1.13×
0.90×
0.29×
0.28×
AWQ-INT4 (vLLM)
FP8 (vLLM)
NF4 (HF)
LLM.int8() (HF)
(b) MMLU change vs BF16
MMLU change (points)
−1.6
−0.7
−0.1
−0.7
−0.6
−0.3
−0.2
+0.7
(c) Qwen3 MMLU vs. memory
MMLU accuracy (%)
Qwen3-32B
Qwen3-8B
HF BF16
HF LLM.int8()
HF NF4
vLLM FP8
vLLM AWQ-INT4
does not fit in 32 GB
32B BF16 needs 66 GB
weight memory (GB, log scale)